

Family Name _____

First Name _____

Student Number

Venue _____

Seat Number _____



No electronic/communication devices are permitted.

No exam materials may be removed from the exam room.

Civil and Natural Resources Engineering EXAMINATION

Mid-Year Examinations, 2025

ENCI335-25S1 (C) Structural Analysis and Systems 1

For Examiner Use Only

Question Mark

Examination Duration: 150 minutes

Exam Conditions:

Closed Book: Students may bring in writing implements only, no materials.

Calculators with a 'UC' sticker approved.

No exam materials may be removed from the exam room.

Materials Permitted in the Exam Venue:

None

Materials to be Supplied to Students:

1 x Write-on exam paper

String to tie exam materials together.

Instructions to Students:

- Total marks: 50
- Answer all questions.
- Formula sheet attached.
- Provide answers inside the provided boxed space only.
- If you find the boxed space insufficient, put a note, continue your answers on additional sheets, and attach them to the back of the exam book.
- All numbers provided have consistent units.

Total _____

Questions Start on Page 3

Question 1. (Total = 10 marks)

Calculate the degree of static indeterminacy (SI), and provide a **qualitative** sketch of the possible deflected shape, bending moment diagram, and shear force diagram for the structures shown in Figures 1 and 2.

a) [5 marks]



Figure 1: The structural diagram for Question 1(a).

[Answer]

[Answer]

[Answer]

b) [5 marks]

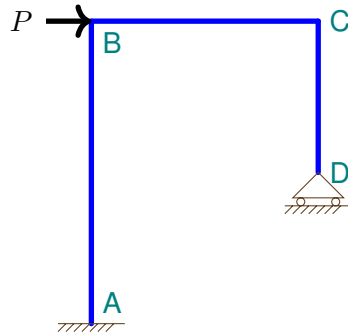


Figure 2: The structural diagram for Question 1(b).

[Answer]

[Answer]

[Answer]

Question 2. (Total = 20 marks)

Consider the following beam (shown in Figure 3) with uniform flexural rigidity EI , and neglect all horizontal force components.

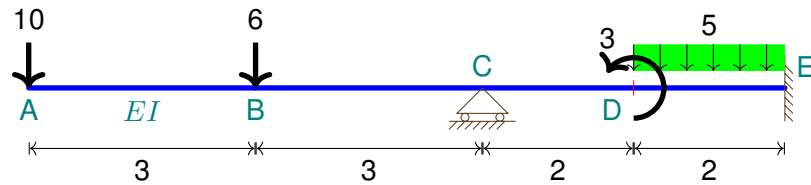


Figure 3: The structural diagram for Questions 2(a) and 2(b).

- a) **[3 marks]** Calculate the degree of static indeterminacy (SI), and specify the displacement and force boundary conditions — that is, identify any known values of translational displacements (v), rotations (θ), shear forces (V), and bending moments (M).

[Answer]

[Answer]

- b) **[6 marks]** Develop the equation for the bending moment $M(x)$ as a function of position x , where $x = 0$ corresponds to the left end of the beam, using the **method of integration**. Use Macaulay brackets where appropriate. Label any unknown redundant reactions using R (for forces) or M (for moments), with subscripts to indicate their locations. Determine the integration constants by applying the force boundary conditions, specifically the known values of shear force (V) and bending moment (M). **Do not proceed to integrate further to obtain the rotation $\theta(x)$ or deflection $v(x)$ functions.**

[Answer]

[Answer]

[Answer]

- c) Suppose the displacement boundary conditions and loads have changed (see Figure 4) such that

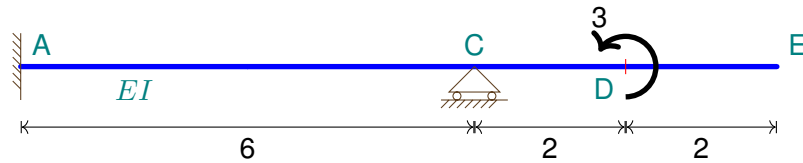


Figure 4: The structural diagram for Question 2(c).

[11 marks] Apply the **Moment-Area Theorems** and the **Principle of Superposition** to determine all reaction forces and moments in the structure. **No marks will be awarded for using the table of beam deflections.** Then, draw the deflected shape, the bending moment diagram, and the shear force diagram. Clearly indicate the locations and values of all maximum and minimum points on each diagram.

[Answer]

[Answer]

[Answer]

[Answer]

Question 3. (Total = 20 marks)

Consider the following portal frame shown in Figure 5, assuming a uniform flexural rigidity EI throughout. Neglect axial and shear deformations in all members.

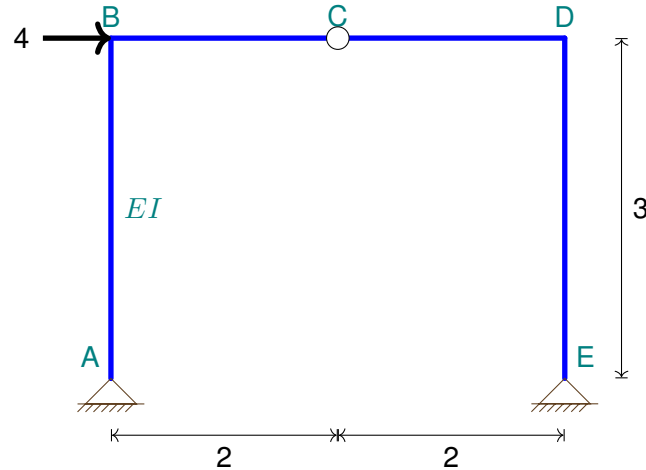


Figure 5: The structural diagram for Question 3.

- a) **[5 marks]** Using the equations of static equilibrium, determine all support reactions. Then, draw the bending moment diagram, clearly indicating the locations and magnitudes of all maximum and minimum values.

[Answer]

[Answer]

[Answer]

- b) **[5 marks]** Apply the **Work-Energy Principle** to calculate the horizontal displacement at point B. **Can this principle be used to determine the rotation at point B? Justify your answer.**

[Answer]

[Answer]

- c) **[2 marks]** Suppose the horizontal displacement at point B is $\frac{12}{EI}$ when the portal frame is subjected to a clockwise point moment of 12 units at B **alone** (i.e., in the absence of any horizontal load at B). Can you use this information to determine the rotation at B **for the load case described in Figure 5**? If so, what is its value?

[Answer]

- d) **[8 marks]** Use the **Unit-Load Method** to calculate the vertical deflection at point C.

[Answer]

[Answer]

[Answer]

Extra Answer Sheets

[Answer]

[Answer]

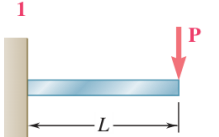
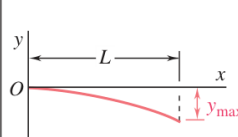
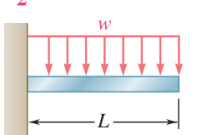
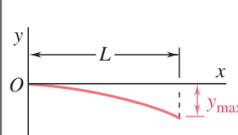
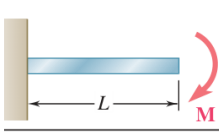
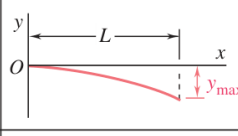
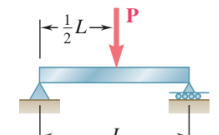
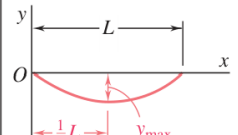
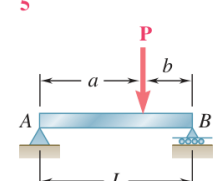
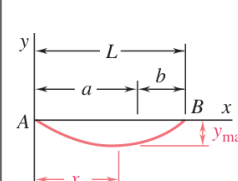
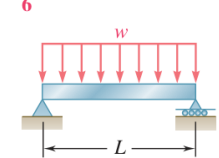
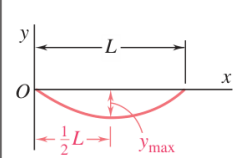
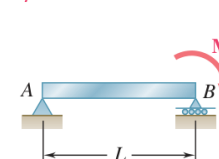
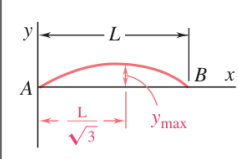
[Answer]

Formulas and Tables

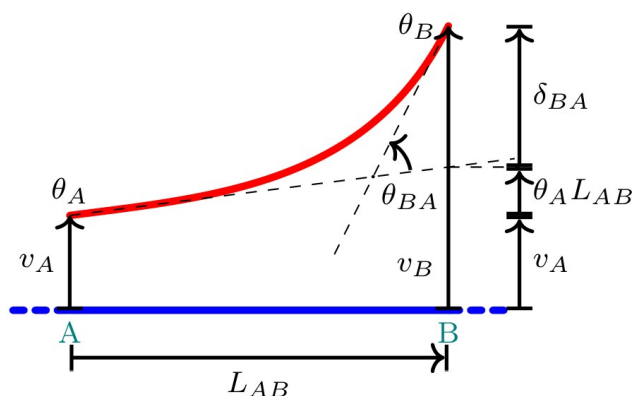
Degree of Static Indeterminacy

$$SI = n_R + n_{MF} - n_E - n_H$$

Table of Beam Deflections

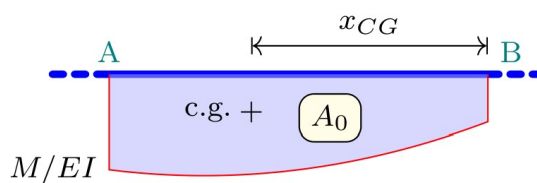
Beam and Loading	Elastic Curve	Maximum Deflection	Slope at End	Equation of Elastic Curve
1 		$-\frac{PL^3}{3EI}$	$-\frac{PL^2}{2EI}$	$y = \frac{P}{6EI}(x^3 - 3Lx^2)$
2 		$-\frac{wL^4}{8EI}$	$-\frac{wL^3}{6EI}$	$y = -\frac{w}{24EI}(x^4 - 4Lx^3 + 6L^2x^2)$
3 		$-\frac{ML^2}{2EI}$	$-\frac{ML}{EI}$	$y = -\frac{M}{2EI}x^2$
4 		$-\frac{PL^3}{48EI}$	$\pm \frac{PL^2}{16EI}$	For $x \leq \frac{1}{2}L$: $y = \frac{P}{48EI}(4x^3 - 3L^2x)$
5 		For $a > b$: $-\frac{Pb(L^2 - b^2)^{3/2}}{9\sqrt{3}EIL}$ at $x_m = \sqrt{\frac{L^2 - b^2}{3}}$	$\theta_A = -\frac{Pb(L^2 - b^2)}{6EIL}$ $\theta_B = +\frac{Pa(L^2 - a^2)}{6EIL}$	For $x < a$: $y = \frac{Pb}{6EIL}[x^3 - (L^2 - b^2)x]$ For $x = a$: $y = -\frac{Pa^2b^2}{3EIL}$
6 		$-\frac{5wL^4}{384EI}$	$\pm \frac{wL^3}{24EI}$	$y = -\frac{w}{24EI}(x^4 - 2Lx^3 + L^3x)$
7 		$\frac{ML^2}{9\sqrt{3}EI}$	$\theta_A = +\frac{ML}{6EI}$ $\theta_B = -\frac{ML}{3EI}$	$y = -\frac{M}{6EIL}(x^3 - L^2x)$

Moment-Area Theorems



$$v_B = v_A + \theta_A L_{AB} + \delta_{BA}$$

Relates the deflections at A & B



1st Moment-Area Theorem:

$$\theta_{BA} = \theta_B - \theta_A = \int_{L_{AB}} \frac{M}{EI} dx = A_0$$

2nd Moment-Area Theorem:

$$\delta_{BA} = \int_{L_{AB}} \frac{M}{EI} x'_B dx = A_0 \cdot x_{CG}$$

The change in θ from A to B:

θ_{BA} = the M/EI area between A & B

The tangential deviation of B wrt. A:

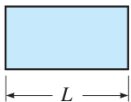
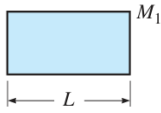
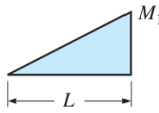
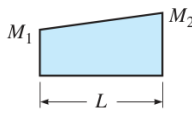
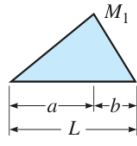
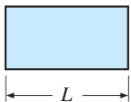
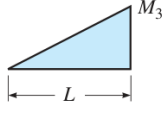
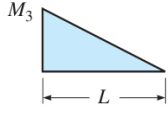
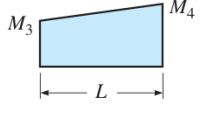
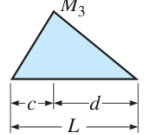
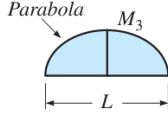
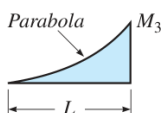
δ_{BA} = the first moment of the M/EI area about B

Shape		Area	c
Rectangle		bh	$\frac{b}{2}$
Triangle		$\frac{bh}{2}$	$\frac{b}{3}$
Parabolic spandrel		$\frac{bh}{3}$	$\frac{b}{4}$

Cubic spandrel		$\frac{bh}{4}$	$\frac{b}{5}$
General spandrel		$\frac{bh}{n+1}$	$\frac{b}{n+2}$

Energy

Table A.2: Values of Product Integrals $\int_{x=0}^{x=L} M_Q M_P dx$

$M_P \backslash M_Q$ 				
	$M_1 M_3 L$	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{2} (M_1 + M_2) M_3 L$	$\frac{1}{2} M_1 M_3 L$
	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{3} M_1 M_3 L$	$\frac{1}{6} (M_1 + 2M_2) M_3 L$	$\frac{1}{6} M_1 M_3 (L + a)$
	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{6} M_1 M_3 L$	$\frac{1}{6} (2M_1 + M_2) M_3 L$	$\frac{1}{6} M_1 M_3 (L + b)$
	$\frac{1}{2} M_1 (M_3 + M_4) L$	$\frac{1}{6} M_1 (M_3 + 2M_4) L$	$\frac{1}{6} M_1 (2M_3 + M_4) L$ $+ \frac{1}{6} M_2 (M_3 + 2M_4) L$	$\frac{1}{6} M_1 M_3 (L + b)$ $+ \frac{1}{6} M_1 M_4 (L + a)$
	$\frac{1}{2} M_1 M_3 L$	$\frac{1}{6} M_1 M_3 (L + c)$	$\frac{1}{6} M_1 M_3 (L + d)$ $+ \frac{1}{6} M_2 M_3 (L + c)$	for $c \leq a$: $\left(\frac{1}{3} - \frac{(a-c)^2}{6ad} \right) M_1 M_3 L$
	$\frac{2}{3} M_1 M_3 L$	$\frac{1}{3} M_1 M_3 L$	$\frac{1}{3} (M_1 + M_2) M_3 L$	$\frac{1}{3} M_1 M_3 \left(L + \frac{ab}{L} \right)$
	$\frac{1}{3} M_1 M_3 L$	$\frac{1}{4} M_1 M_3 L$	$\frac{1}{12} (M_1 + 3M_2) M_3 L$	$\frac{1}{12} M_1 M_3 \left(3a + \frac{a^2}{L} \right)$

End of Examination